

Reliable Evaluation of ARIMA-LSTM Hybrids for Remaining Useful Life Prediction: A Multi-Seed, Specification-Sensitive Study on NASA C-MAPSS

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Abstract: Adding an ARIMA trend feature to an LSTM is a common design for remaining useful life (RUL) prediction. We evaluate it on NASA C-MAPSS rather than proposing a new model, using 20 seeds, three ARIMA specifications (fixed order, AIC-selected order, and a causal expanding-window construction), two cheap smoothers, and two capacity controls. Effects are dataset- and protocol-dependent. On the fixed test set, AIC-selected (non-causal) ARIMA lowered mean RMSE on FD002 by about 1.0 cycle (seed-level Holm $p = 0.0004$), but none of the FD002 effects survived a two-level test that also resamples test engines; on FD001 and FD003 trend features did not help. A duplicated-input control with no new information reproduces an FD004 penalty of similar size to the ARIMA hybrids', suggesting input width contributes substantially. Causal and fixed-order hybrids did not differ beyond a ± 1 -cycle margin, and a moving average had lower RMSE than causal ARIMA on three of four datasets at the seed level, at about 1/2000 of the feature cost. Few-run conclusions are fragile: six seeds gave the opposite sign to the 20-seed estimate in 6 of 20 comparisons, and 15 of 48 comparison cells are significant at the seed level but only 2 under the two-level test.

Index Terms: remaining useful life, predictive maintenance, LSTM, time-series forecasting, feature engineering, industrial IoT, NASA C-MAPSS, reproducibility

I. INTRODUCTION

Estimating remaining useful life (RUL) from streaming sensor data is central to prognostics and health management (PHM) [1], and a recent survey [10] catalogs the deep-learning approaches now used for it. NASA's C-MAPSS turbofan simulation [1] is the standard benchmark: a controlled, multi-regime, multi-fault-mode setting where model-comparison claims can in principle be checked. A common design fits an ARIMA model per sensor channel and concatenates its trend with the raw signal before an LSTM regressor [17], letting the network spend capacity on residual nonlinear dynamics.

But deep models are trained stochastically, and an ARIMA feature is itself a specification choice: order, fitting window, and causality with respect to the prediction target. To our knowledge, neither seed robustness nor the causal validity of the ARIMA feature has been examined for ARIMA-LSTM RUL hybrids [9]. We therefore evaluate, rather than propose,

ARIMA trend augmentation: all four C-MAPSS sub-datasets, 20 seeds, three ARIMA specifications, cheap alternatives, and controls that separate the information a trend feature carries from the extra input width it brings.

Research questions. RQ1: does an ARIMA trend feature help an LSTM, and does causal construction matter? RQ2: how reliable are conclusions drawn from few seeds? RQ3: how much does ARIMA order specification matter? RQ4: is any effect due to trend information or to input width and capacity? RQ5: can a cheap, fitting-free smoother replace ARIMA?

Contributions: (1) a 20-seed, four-dataset evaluation with matched protocol; (2) a causal ARIMA construction that uses no observation after the current cycle and did not differ from the standard construction beyond a ± 1 -cycle margin; (3) capacity controls (parameter-matched and duplicated-input LSTMs), one of which reproduces an FD004 penalty of similar size to the ARIMA hybrids', indicating that input width likely contributes substantially; (4) evidence that a moving average had lower RMSE than causal ARIMA on three of four datasets at the seed level, at a tiny fraction of the cost; (5) a quantification of how few-seed studies mislead, with seed-level and two-level (seeds and engines) tests under Holm-Bonferroni correction. Code, per-seed predictions, and pinned package versions are public at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>.

II. RELATED WORK

RUL on C-MAPSS. The benchmark [1] underlies most data-driven RUL work; Ramasso and Saxena [2] laid out early benchmarking protocols, early deep models used CNNs [20] and LSTMs [21], and recent architectures include CNN-LSTM [5] and residual-corrected BiLSTM [6] hybrids and an uncertainty-aware Inception-BiLSTM-attention model [8]. Su and Fang [7] decompose degradation into trend, operational-state, and residual components with a learned embedding, which motivates our trend/residual split.

Hybrid statistical/neural forecasting. ARIMA-neural hybrids have a long history [17] and have been applied to RUL-adjacent problems [9], without, to our knowledge, examining

seed robustness or the causal validity of the ARIMA construction.

Reproducibility and seed sensitivity. Single-seed deep reinforcement learning results routinely fail to replicate [11], and exact reproducibility does not imply robustness to small setup changes [12]; a companion paper [13] proposes concrete variance-estimation protocols. Colas et al. [22] show how the number of seeds governs the error rates of significance tests, and Agarwal et al. [23] argue that few-run evaluations must report interval estimates rather than point estimates. Picard [14] shows that outlier seeds can perform much better or worse than average in image classification, and Jordan [15] traces test-set variance in classifiers to a calibration property of the trained network. Demšar [16] sets out the paired, multiple-comparison-corrected testing we follow, with Holm’s step-down procedure [24], and Lakens [25] describes the equivalence testing we use to make claims of no practical difference. A contemporaneous benchmark on this dataset [19] uses one fixed seed, and a parallel audit [26] finds preprocessing choices materially affect reported C-MAPSS metrics. We treat ARIMA specification, causal construction, and seed count as experimental variables.

III. METHODOLOGY

Data and preprocessing. All four C-MAPSS sub-datasets [1]: FD001/FD003 (100 engines, 1 condition, 1/2 fault modes) and FD002/FD004 (260/249 engines, 6 conditions, 1/2 fault modes). RUL is capped at 125 cycles and near-constant sensors are dropped (15 of 21 retained for FD001/FD003). For FD002/FD004 we K -means cluster the operating settings into 6 regimes (fit on training data only) and z -score per regime [4], clipped to ± 6 standard deviations.

ARIMA trend features. For each (unit, sensor) the fitted trend $\hat{y}_t$ is concatenated with the raw features before a 2-layer LSTM (hidden size 64) [3] and a $64 \rightarrow 32 \rightarrow 1$ head. *Fixed order:* ARIMA(1,1,0), parameters fit once by MLE on the unit’s whole available series. *AIC-selected:* the best of (1,1,0), (0,1,1), (1,1,1), (2,1,0) under the same full-series fit; like the fixed order it is non-causal, and it isolates order from causality. *Causal:* ARIMA(1,1,0) (statsmodels, no constant) is first fit by MLE on cycles 1 to 10, and its in-sample one-step predictions serve as the trend for those ten cycles. For each later cycle t : if at least 20 cycles have passed since the last refit, the model is first re-estimated by full MLE on cycles 1.. t (so at these refit cycles, one in 20, the parameters used at t include y_t itself); otherwise the Kalman state of the current fit is updated with y_t and the parameters are left unchanged (append, no refit). The trend feature at t is then the model’s one-step-ahead prediction of y_t , computed from the filtered state through cycle $t - 1$. Every feature is thus a function of cycles 1.. t only: no observation after cycle t is used, so the construction is causal. At refit cycles it is not strictly ex ante, because y_t , which the LSTM also receives as a raw input at t , contributes to the parameters. Two bounded exceptions apply: the first ten cycles use parameters estimated on cycles up to 10, and series of 11 or fewer cycles, or fits that

raise an exception, fall back to a trailing 5-cycle mean (none occurred in 27,536 series). The fixed and AIC variants fit each unit’s whole available series at once (complete run-to-failure for training engines, cycles up to the prediction point for test engines), so they are non-causal.

Controls and baselines. A *moving average* (trailing 10 cycles) and *exponential smoothing* ($\alpha = 0.3$) replace ARIMA with trivially causal, fitting-free features. Two plain-LSTM controls separate information from capacity: a *parameter-matched* LSTM (hidden size 66 or 67, matching the hybrid’s parameter count) and a *duplicated-input* LSTM (raw features concatenated with an exact copy, as wide as the hybrid but with no new information). A CNN-LSTM [5] checks the backbone. All models use Adam (10^{-3}), MSE loss on capped RUL, window length 30, and early stopping on 15% held-out validation engines (patience 8, at most 60 epochs). Each test engine is scored once from its final window; we report RMSE, MAE, and the PHM score [1].

IV. EXPERIMENTAL SETUP

All primary comparisons use 20 seeds unless otherwise stated: the original six (42, 123, 2024, 7, 99, 314) and 14 more fixed in advance (1 to 15 except 7); CNN-LSTM, gated fusion, and the window-length ablation use the original six seeds. With PyTorch 2.14.0 on CPU, reruns reproduce archived results (main pipeline on FD001 to four decimals; all 24 AIC-selected reruns to within 4×10^{-15}). ARIMA features depend only on the normalised series, not the seed, so they are computed once per dataset and reused; a SHA-256 fingerprint of the normalised data guards each reuse. About 6.5% to 7.0% of ARIMA fits (fixed, AIC, causal) did not fully converge; such fits are kept, and only exceptions trigger the fallback. A seed fixes model initialisation, mini-batch order, dropout masks, the K -means seed, and the validation-engine split (15% of training engines, drawn from a generator seeded with the run’s seed). Seed-to-seed differences therefore combine training randomness with validation-split variation, which we call run-level variation. The split is identical across all models under one seed, so paired comparisons share it.

Statistical protocol. The seed-level test is a two-sided paired Wilcoxon signed-rank test on the 20 per-seed test RMSEs of two models; it asks whether the two procedures differ across runs on this fixed test set. The two-level test uses the $S \times E$ table of paired per-engine absolute-error differences ($S = 20$ seeds, $E = 100$ to 259 test engines), treats seeds and engines as crossed random factors, and takes the grand mean difference as its statistic. We compute (i) a variance-components t -test, with $\text{Var} = \hat{\sigma}_{\text{seed}}^2 / S + \text{MS}_{\text{eng}} / (SE)$ (seed variance floored at zero) and Satterthwaite degrees of freedom, and (ii) a two-sided pigeonhole bootstrap [18] with 40,000 replicates, each resampling seeds and engines independently with replacement and recomputing the grand mean. We keep the larger p , since the pigeonhole bootstrap is conservative and the variance-components estimator can be optimistic with few clusters; 90% intervals are the union of the t interval and the bootstrap 5th to 95th percentiles. Both tests use Holm-Bonferroni correction

TABLE I

TEST RMSE (CYCLES), MEAN $\pm$ STD OVER 20 SEEDS. BEST 20-SEED MEAN PER DATASET IN BOLD; CNN-LSTM HAS SIX SEEDS ONLY.

Model	FD001	FD002	FD003	FD004
Plain LSTM	14.26 $\pm$ 0.64	14.75 $\pm$ 0.62	14.36 $\pm$ 1.30	15.05 $\pm$ 0.29
Fixed-order ARIMA	14.63 $\pm$ 1.15	14.39 $\pm$ 0.53	14.99 $\pm$ 0.72	15.52 $\pm$ 0.48
AIC-selected ARIMA	15.00 $\pm$ 1.00	13.75$\pm$0.79	14.61 $\pm$ 0.86	14.75$\pm$0.56
Causal ARIMA	14.81 $\pm$ 0.88	14.44 $\pm$ 0.46	14.56 $\pm$ 0.67	15.48 $\pm$ 0.45
Moving average	14.25$\pm$0.72	14.06 $\pm$ 0.43	14.84 $\pm$ 0.71	14.75 $\pm$ 0.58
Exp. smoothing	14.56 $\pm$ 0.98	14.23 $\pm$ 0.61	14.76 $\pm$ 0.96	15.12 $\pm$ 0.47
Param-matched plain	14.54 $\pm$ 0.70	14.62 $\pm$ 0.47	14.31$\pm$0.74	15.08 $\pm$ 0.50
Dup.-input plain	14.52 $\pm$ 0.82	14.85 $\pm$ 0.55	14.81 $\pm$ 1.01	15.55 $\pm$ 0.49
CNN-LSTM (6 seeds)	14.47 $\pm$ 0.59	15.11 $\pm$ 1.05	13.90 $\pm$ 0.70	14.86 $\pm$ 0.54

TABLE II

MEAN PER-SEED RMSE DIFFERENCE (FIRST MINUS SECOND, CYCLES; NEGATIVE FAVOURS THE FIRST). *: SEED-LEVEL HOLM $p < 0.05$ (RMSE); †: TWO-LEVEL HOLM $p < 0.05$ (PER-ENGINE ABSOLUTE ERROR).

Comparison	FD001	FD002	FD003	FD004
Fixed ARIMA – plain	+0.37	−0.36	+0.64	+0.48*
AIC ARIMA – plain	+0.74*	−1.00*	+0.25	−0.30
Causal ARIMA – plain	+0.55	−0.31	+0.21	+0.44*
Moving avg – plain	−0.01	−0.69*	+0.48	−0.29
Exp. smooth – plain	+0.30	−0.52*	+0.41	+0.08
Param-matched – plain	+0.28	−0.13	−0.05	+0.03
Dup. input – plain	+0.26	+0.10	+0.46	+0.50*
AIC – fixed ARIMA	+0.37	−0.64*	−0.39	−0.78*†
Causal – fixed ARIMA	+0.18	+0.05	−0.43	−0.04
Moving avg – causal	−0.56*	−0.38*	+0.27	−0.73*
Fixed ARIMA – dup. input	+0.11	−0.46*	+0.18	−0.02
Moving avg – dup. input	−0.27	−0.79*	+0.02	−0.80*†

across the four datasets within each comparison. Equivalence uses two one-sided tests against a prespecified margin of ± 1 cycle (about 10% of MAE), Holm-adjusted. The two-level and equivalence analyses use per-engine absolute error (MAE scale); the seed-level test and Table II use RMSE.

V. RESULTS

A. RQ1: Do Trend Features Help, and Does Causality Matter?

Table I and Fig. 1 show no uniform winner. The plain LSTM is best or tied on FD001 and FD003; on FD002, every trend-augmented hybrid has lower RMSE than plain, by 0.3 to 1.0 cycles, and the AIC-selected hybrid is lowest (13.75 vs. 14.75; seed-level Holm $p = 0.0004$, Table II). None of the FD002 differences survives the two-level test (all Holm $p \geq 0.31$, Table VI). On FD001 the AIC hybrid is worse (+0.74, seed-level $p = 0.041$), and on FD004 the fixed-order and causal hybrids are worse (+0.48, $p = 0.008$; +0.44, $p = 0.003$), again at the seed level only.

Causality did not cost accuracy: the causal and fixed-order hybrids differ by at most 0.5 cycles of per-engine absolute error, and on all four datasets the 90% interval lies within ± 1 cycle with Holm-adjusted TOST $p \leq 0.023$ (Table III), so they did not differ beyond our prespecified margin. Its cost is compute: about 222 ms of CPU per series against 5.1 ms for the full-series fit (FD002; 274 vs. 6.3 ms on FD001).

TABLE III

LARGEST ABSOLUTE ENDPOINT OF THE TWO-LEVEL 90% INTERVAL FOR THE MEAN PAIRED DIFFERENCE IN PER-ENGINE ABSOLUTE ERROR (CYCLES). BOLD: WITHIN THE PRESPECIFIED ± 1 -CYCLE MARGIN. †: EQUIVALENT BY HOLM-ADJUSTED TOST ($p < 0.05$).

Comparison	FD001	FD002	FD003	FD004
Fixed ARIMA – plain	0.84†	0.57†	1.07	0.59†
AIC ARIMA – plain	1.36	1.43	0.94	1.16
Causal ARIMA – plain	0.95†	0.43†	0.57†	0.57†
Moving avg – plain	0.84	0.93	0.97	0.66†
Exp. smooth – plain	0.78†	0.73†	0.85†	0.38†
Param-matched – plain	0.58†	0.39†	0.60†	0.26†
Dup. input – plain	0.71†	0.61†	0.98†	0.64†
Causal – fixed ARIMA	0.57†	0.51†	0.91†	0.27†
Moving avg – causal	1.16	0.94	0.81†	0.89

TABLE IV

PROBABILITY THAT k RANDOM SEEDS RECOVER THE SIGN OF THE 20-SEED MEAN RMSE DIFFERENCE (MEDIAN / WORST CELL), BY EFFECT SIZE $|\Delta|$ (CYCLES). LAST ROW: MEAN PROBABILITY THAT SOME CELL IS TWO-LEVEL SIGNIFICANT.

	$k=3$	$k=6$	$k=10$
All 20 cells	0.80 / 0.50	0.89 / 0.51	0.98 / 0.54
$ \Delta \geq 0.5$ (7 cells)	0.88 / 0.81	0.97 / 0.91	1.00 / 0.98
$ \Delta < 0.3$ (6 cells)	0.69 / 0.50	0.75 / 0.51	0.86 / 0.54
Any two-level significant cell (mean)	0.12	0.10	0.11

B. RQ2: How Reliable Are Few-Seed Conclusions?

Relative to the 20-seed estimate, our first six seeds gave the opposite sign of the mean RMSE difference in 6 of the 20 dataset-and-comparison cells we examined (five comparisons, four datasets). The fixed-order hybrid on FD002 looked worse than plain at six seeds (+0.34) and is better at 20 (−0.36); the AIC hybrid on FD001 flips from −0.05 to +0.74. For a random six-seed study, the chance of recovering the 20-seed sign ranges from 0.51 to 1.00 across these cells (median 0.89). Recovery depends on effect size (Table IV): for the seven cells with $|\Delta| \geq 0.5$ cycles the sign is recovered with probability at least 0.91 at six seeds, whereas for the six cells with $|\Delta| < 0.3$ cycles the median is 0.75 at six seeds and 0.86 at ten, and a near-zero effect (moving average on FD001) stays at 0.54. Fig. 2 shows the running estimate settling only after roughly ten seeds.

Engine-level tests on few seeds are also over-confident, because they treat the trained networks as fixed. A six-seed engine-level test (paired Wilcoxon on seed-averaged per-engine errors) flags the fixed-order penalty on FD004 (raw $p = 0.004$); under the two-level test it is not significant at six or at 20 seeds. *Where the uncertainty comes from.* Across the 48 comparison cells of Table II, run-level (seed) variation accounts for a median of 14% of the variance of the mean per-engine error difference (interquartile range 7% to 30%); the median standard error is 0.09 cycles from seeds alone and 0.20 cycles from test-engine sampling. With 20 seeds, in the median cell most of the remaining uncertainty is about which engines are tested, consistent with the frequent

TABLE V
SECONDARY METRICS OVER 20 SEEDS: MAE (CYCLES, MEAN $\pm$ STD) AND MEAN PHM SCORE (LOWER IS BETTER).

Model	MAE				PHM score			
	FD001	FD002	FD003	FD004	FD001	FD002	FD003	FD004
Plain LSTM	10.48 $\pm$ 0.52	10.35 $\pm$ 0.37	9.91 $\pm$ 0.90	10.69 $\pm$ 0.32	431	1282	581	1178
Fixed ARIMA	10.78 $\pm$ 0.78	10.17 $\pm$ 0.38	10.44 $\pm$ 0.47	10.92 $\pm$ 0.35	454	1267	722	1351
AIC ARIMA	10.82 $\pm$ 0.76	9.64 $\pm$ 0.47	9.70 $\pm$ 0.60	10.07 $\pm$ 0.41	423	1029	876	1265
Causal ARIMA	10.89 $\pm$ 0.70	10.39 $\pm$ 0.49	9.95 $\pm$ 0.42	10.92 $\pm$ 0.39	495	1212	717	1213
Moving avg	10.40 $\pm$ 0.64	9.93 $\pm$ 0.29	10.25 $\pm$ 0.44	10.43 $\pm$ 0.45	415	1089	708	1094

TABLE VI
HOLM-ADJUSTED p -VALUES FOR EVERY COMPARISON IN TABLE II: SEED-LEVEL TEST / TWO-LEVEL TEST. BOLD: $p < 0.05$.

Comparison	FD001	FD002	FD003	FD004
Fixed ARIMA – plain	0.312 / 0.746	0.098 / 0.746	0.098 / 0.423	0.008 / 0.746
AIC ARIMA – plain	0.041 / 1.000	<0.001 / 0.313	0.498 / 1.000	0.097 / 0.289
Causal ARIMA – plain	0.064 / 0.788	0.330 / 1.000	0.452 / 1.000	0.003 / 0.788
Moving avg – plain	0.869 / 0.873	0.007 / 0.664	0.316 / 0.873	0.316 / 0.873
Exp. smooth – plain	0.624 / 1.000	0.017 / 0.549	0.269 / 0.958	0.624 / 1.000
Param-matched – plain	0.422 / 1.000	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000
Dup. input – plain	0.783 / 0.494	0.783 / 0.489	0.783 / 0.489	0.006 / 0.152
AIC – fixed ARIMA	0.211 / 0.943	0.025 / 0.403	0.211 / 0.192	<0.001 / 0.005
Causal – fixed ARIMA	1.000 / 1.000	1.000 / 0.538	0.069 / 0.227	1.000 / 1.000
Moving avg – causal	0.038 / 0.451	0.028 / 0.349	0.070 / 0.451	<0.001 / 0.189
Fixed ARIMA – dup. input	1.000 / 1.000	0.038 / 0.099	1.000 / 1.000	1.000 / 1.000
Moving avg – dup. input	0.165 / 0.979	<0.001 / 0.052	0.841 / 0.979	<0.001 / 0.038

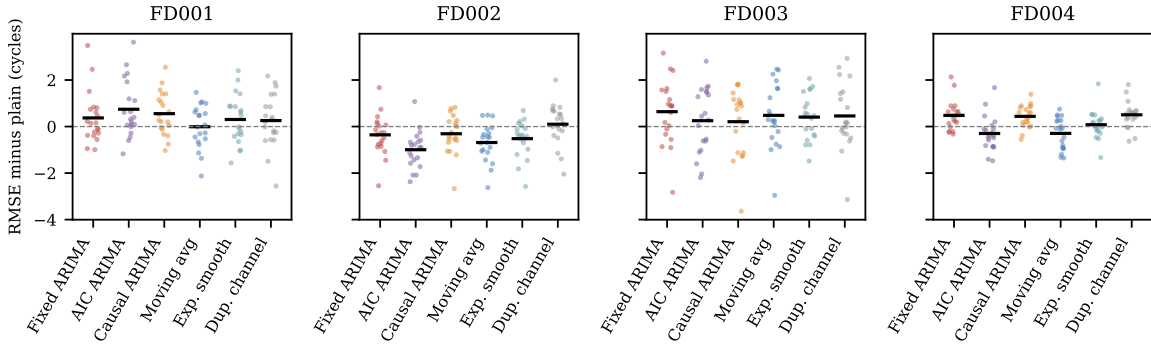


Fig. 1. Per-seed test RMSE of each variant minus the plain LSTM (20 seeds; bars mark the mean). Trend features lowered RMSE on FD002 but not on FD001 or FD003, and the duplicated-input control shows a penalty of similar size to the ARIMA hybrids’ on FD004.

disagreement between the seed-level and two-level tests: more seeds sharpen the former, but claims about other engines are limited by the 100 to 259 test engines. *Power*: For the fixed, causal and AIC hybrids against plain and the moving average against causal ARIMA (four datasets each), we solved a two-sided paired t -test on the 20 per-seed RMSE differences for 80% power at $\alpha = 0.05$, using the observed mean and standard deviation (effect size $|\bar{d}|/s_d$, noncentral t); detecting the observed differences would need 7 to 380 seeds, and a half-cycle difference (mean set to 0.5 cycles) 10 to 67. Both inputs are estimated from 20 seeds, and effects selected for being large are likely overestimated, so these figures are approximate.

C. RQ3: Order Specification

The AIC-selected order has lower RMSE than the fixed order on FD004 (-0.78 ; seed-level $p = 0.0008$) and FD002 (-0.64 ; $p = 0.025$), and FD004 is the one cell in this family that also survives the two-level test (Holm $p = 0.005$). Order selection overwhelmingly preferred $(0, 1, 1)$ (76% to 82% of series). Because AIC selection is non-causal, this is a specification-sensitivity result under the original protocol; it is not evidence about how a causally fit AIC model would behave.

D. RQ4: Trend Information or Input Width?

A parameter-matched plain LSTM did not differ from plain by more than 0.6 cycles on any dataset (Table III), so parameter count alone did not change accuracy detectably.

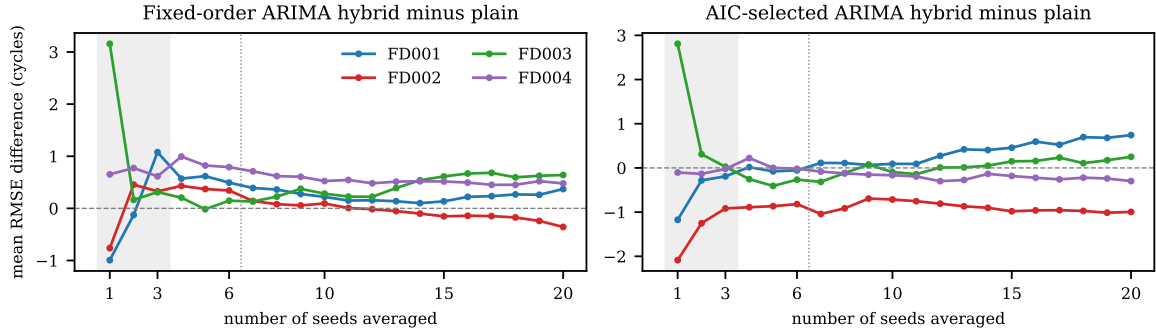


Fig. 2. Running mean RMSE difference (hybrid minus plain) as seeds are added in the order used. The shaded region is a three-seed study, the dotted line the original six seeds.

The duplicated-input control, which adds no information, shows a penalty on FD004 of similar size to the ARIMA hybrids’ (+0.50 RMSE, against +0.48 for the fixed-order and +0.44 for the causal hybrid; seed-level $p = 0.006$, two-level $p = 0.15$), and the fixed-order hybrid does not differ from it (−0.02). This indicates that input width likely contributes substantially to the FD004 degradation, rather than the penalty uniquely implicating the ARIMA trend feature; it does not show that width is the only cause. On FD002 the duplicated-input model shows no gain (+0.10), while the moving average is 0.79 lower (seed-level $p = 0.001$, two-level $p = 0.052$), and on FD004 0.80 lower (two-level $p = 0.038$). This is consistent with trend information, not width, driving the FD002 gain, although the FD002 two-level test is borderline.

E. RQ5: Can a Cheap Smoother Replace ARIMA?

The moving average had lower RMSE than causal ARIMA on three of four datasets (−0.56, −0.38, −0.73 on FD001, FD002, FD004; seed-level Holm $p \leq 0.04$) and cannot be distinguished from AIC-selected ARIMA on any dataset; none of the moving-average-versus-causal differences survives the two-level test. Exponential smoothing lowers RMSE on FD002 (−0.52) but not elsewhere. The moving average takes 0.10 ms per series, roughly $2,000\times$ less than causal ARIMA, and neither smoother improves on plain on FD003.

What survives the two-level test. Table VI lists both p -values for every cell. Of 48 comparison cells in Table II, 15 are significant at the seed level but only 2 at the two-level: AIC vs. fixed order on FD004 (Holm $p = 0.005$) and the moving average vs. the duplicated-input control on FD004 ($p = 0.038$). The AIC hybrid also has lower RMSE than the duplicated-input control on FD004 (two-level $p = 0.030$). MAE follows RMSE (Table V); the PHM score, which penalises late predictions, agrees on FD002 (AIC ARIMA lowest) but not on FD004, where the AIC hybrid’s lower RMSE and MAE come with a higher PHM score than plain (1265 vs. 1178), a reminder that the metrics weight errors differently. Table III gives the interval endpoints. At the prespecified ± 1 -cycle margin, Holm-adjusted TOST establishes equivalence to plain in 20 of the 24 cells for the fixed-order, causal, moving-average, exponential-

smoothing, parameter-matched and duplicated-input variants: all four datasets for the causal hybrid, exponential smoothing and both controls, three for the fixed-order hybrid (its FD003 interval reaches +1.07), and only FD004 for the moving average, whose intervals lie within ± 1 on all four datasets but with Holm-adjusted $p = 0.07$ on the other three. Equivalence to plain is not established for the AIC hybrid on any dataset (endpoints up to 1.43).

F. Additional Checks (Six Seeds)

Three further checks used the original six seeds only, so, given RQ2, they are suggestive. CNN-LSTM has the best six-seed mean RMSE on FD003 (13.90) and the worst on FD002 (15.11), but no difference from plain survives either test; a gated-fusion variant differs significantly from neither plain nor the AIC hybrid; and a window-length ablation on FD001 found a hybrid advantage at $W = 20$ and 30 (0.68 and 0.54 cycles) under a shorter training budget (patience 6, at most 50 epochs), whereas the full-budget 20-seed comparison at $W = 30$ does not favour the hybrid (+0.37, not significant), so the ablation should not be read as protocol-matched.

VI. DISCUSSION

The paper’s value does not depend on ARIMA winning. On this fixed test set, trend features lowered RMSE on FD002, the six-regime single-fault dataset, at the seed level, and the moving average, the cheapest trend feature, obtained about two-thirds of the AIC-selected gain (−0.69 vs. −1.00). None of the FD002 effects survives the two-level test, so they are not established for other engines; the general point is that seed-level and two-level inference can disagree. On FD001 and FD003 trend features did not help. On FD004 a duplicated-input control shows a penalty of similar size to the ARIMA hybrids’, which suggests that input width contributes substantially to the apparent ARIMA penalty. We have no mechanistic account of why FD002 differs. The AIC findings come from the non-causal protocol and are not conclusions about the deployable specification; the causal-ARIMA and smoother results are.

The methodological lesson is the paper’s contribution. With six seeds, 6 of 20 comparisons had the opposite sign to the

20-seed estimate, and 15 of 48 cells that are significant at the seed level shrink to 2 under the two-level test. Table IV suggests how many seeds are enough: for effects of at least half a cycle, six seeds recovered the sign in at least 91% of random draws, but effects below 0.3 cycles were recovered only 75% of the time at six seeds and 86% at ten, and a near-zero effect stayed near 50%. Six seeds should therefore be treated as a minimum reporting standard, ten or more may be needed for effects near one cycle or less, and near-zero effects can require substantially larger budgets (up to about 380 seeds by the power calculation above).

Checklist. Report at least six seeds, more for effects near a cycle or less (Table IV); give effect sizes with intervals; include a capacity-matched control and a cheap non-ARIMA baseline; report seed-level and two-level tests with multiple-comparison correction; and verify that any deployed feature uses no observation after the predicted cycle.

VII. LIMITATIONS

Twenty seeds is moderate power for effects of a fraction of a cycle. All experiments use one simulated benchmark, one recurrent family, and a fixed test set, so generalization to real industrial streams is untested. The AIC results are non-causal, and no causal AIC selection was run. The causal construction is not strictly *ex ante* at refit cycles (one in 20), and a variant that refits on cycles up to $t - 1$ was not run. CNN-LSTM, a gated-fusion variant, and a window-length ablation were run at six seeds only (see the repository). The causal specification costs about 44 times the full-series ARIMA fit, and end-to-end runtimes were measured under concurrent load and are indicative only.

VIII. CONCLUSION

Evaluated over 20 seeds, four datasets, three ARIMA specifications, cheap smoothers, and capacity controls, the value of ARIMA trend augmentation of an LSTM is dataset- and evaluation-protocol dependent. On the fixed test set it lowered RMSE on FD002 at the seed level (most with the non-causal AIC-selected order) but not on FD001 or FD003, and none of the FD002 effects survives a two-level seeds-and-engines test. A duplicated-input control reproduces an FD004 penalty of similar size, so input width likely contributes substantially. The causal construction did not differ from the standard one beyond a ± 1 -cycle margin, and a moving average had lower RMSE than causal ARIMA on three of four datasets at the seed level, at a tiny fraction of the cost. Six seeds gave the opposite sign to the 20-seed estimate in 6 of 20 comparisons, so few-run and seed-level-only results should be treated as provisional, and comparisons should include cheap and capacity-matched baselines. Future work is a second benchmark and backbone, causal AIC selection, and real industrial data.

CODE AND DATA AVAILABILITY

Code, per-seed predictions, and pinned package versions are at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>; the archived results comprise

TABLE VII
FEATURE-CONSTRUCTION CPU TIME PER SERIES (MS).

Dataset	Moving avg	Exp. smoothing	Full-series ARIMA	Causal ARIMA
FD001	0.11	0.06	6.3	274
FD002	0.10	0.05	5.1	222

454 pipeline executions training 1,136 models (four datasets, 20 seeds each for the plain, fixed-order, AIC-selected, causal, moving-average, exponential-smoothing, and capacity-control runs; six seeds for CNN-LSTM and gated fusion; and six seeds of a window-length ablation).

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APPENDIX A ADDITIONAL RESULTS

A. Cost of Feature Construction

Table VII reports single-process CPU time per (unit, sensor) series, with the methods interleaved on the same series so that concurrent machine load affects them equally (252 series on FD002, 180 on FD001; code in the repository). The causal construction costs about 44 times the full-series fit and about 2,000 times the moving average. The AIC-selected variant fits four candidate orders per series. End-to-end sweep runtimes are not comparable across variants, because they include LSTM training and ran under differing concurrent load.

B. ARIMA Convergence

Over all four datasets (train and test series, 27,536 series), 1,783 of 27,536 fixed-order fits (6.5%), 7,244 of 110,144 AIC candidate fits (6.6%), and 18,032 of 258,393 causal fits, including refits (7.0%), did not fully converge. Such fits are kept. No series fell back to the rolling mean.

C. Window-Length Ablation and Gated Fusion (Six Seeds)

Table VIII reports a window-length ablation on FD001 with the fixed-order hybrid, using a shorter training budget (patience 6, at most 50 epochs, 15 validation engines) than the main protocol. Table IX reports a gated-fusion variant, in which a learned per-timestep, per-channel sigmoid gate scales the AIC-selected trend feature. Both were run on the original six seeds only and, given the main results, should be read as suggestive. A paired engine-level Wilcoxon test finds no significant difference between the gated variant and either plain or the AIC hybrid on any dataset (all $p > 0.13$).

TABLE VIII
WINDOW-LENGTH ABLATION, FD001, SIX SEEDS, RMSE (MEAN $\pm$ STD).

W (cycles)	10	20	30	50
Plain	18.60 $\pm$ 0.55	17.39 $\pm$ 0.41	14.69 $\pm$ 0.68	14.46 $\pm$ 0.69
Hybrid	18.62 $\pm$ 0.65	16.71 $\pm$ 0.29	14.15 $\pm$ 0.70	14.63 $\pm$ 1.01

TABLE IX
GATED FUSION, SIX SEEDS, RMSE (MEAN; GATED ALSO $\pm$ STD). THE
FIXED AND AIC COLUMNS ARE THEIR SIX-SEED MEANS.

Dataset	Plain	Fixed	AIC	Gated
FD001	14.43	14.92	14.38	14.97 $\pm$ 0.74
FD002	14.59	14.93	13.77	13.93 $\pm$ 0.35
FD003	14.72	14.87	14.45	14.33 $\pm$ 0.90
FD004	14.92	15.71	14.90	15.03 $\pm$ 0.47

D. Sensor-Trend Linearity

We asked whether FD002's advantage reflects more ARIMA-explainable sensor trends. Fitting ARIMA(0, 1, 1) to 25 training units per dataset and measuring the fraction of each differenced sensor's variance the fitted trend explains, a per-sensor test that treats 364 to 500 unit-sensor pairs as independent found FD002 higher than every other dataset ($p < 0.05$). Sensors within an engine share its degradation trajectory, so the engine is the independent unit. Recomputed on 25 engine-level means per dataset with Holm correction, FD002 remains higher than FD001 (+0.021, $p_{\text{adj}} < 0.001$) and FD003 (+0.040, $p_{\text{adj}} < 0.001$) but not FD004 (+0.007, $p = 0.107$). The FD002 gain is therefore not explained by trend linearity.

E. Reproduction Notes

Reruns of the 24 AIC-selected runs, made to restore per-engine predictions that were not archived originally, matched the previously stored RMSEs to within 4×10^{-15} . All experiments use CPU-only PyTorch 2.14.0 with the versions pinned in the repository.

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